



Costume Design Guild

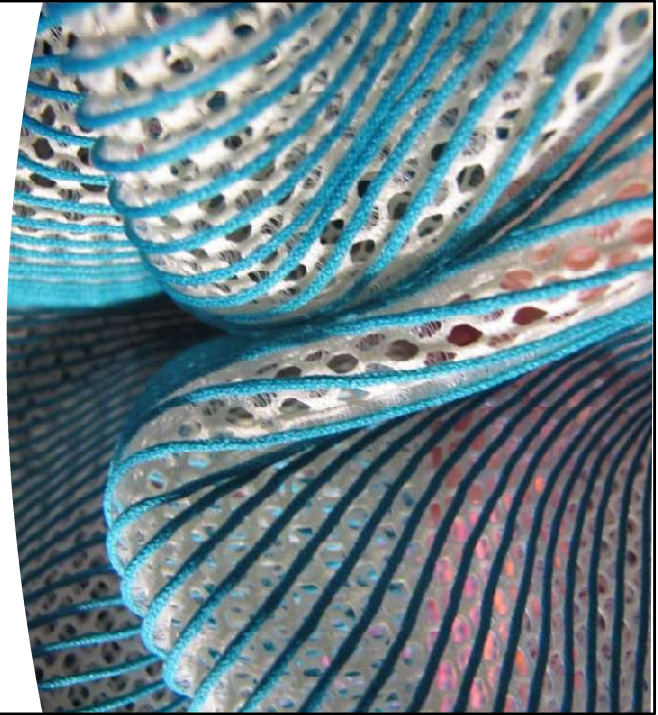
Speculative materials for
extreme environments

Block D

Week 6

Agenda

1. Theory & activity
2. Work review
3. Reflection & wrap-up





Speculative materials

For extreme environments

The 6-step workflow

1. **Identify the purpose:**
What problem does this material solve?
2. **Find real-life base references:**
What exists with similar properties?
3. **Extrapolate:**
Push those properties further
4. **Define the limitations:**
What's the cost, what fails, who can't access it?
5. **Consider visual result:**
Texture, sheen, colour, patterns, tech elements
6. **Think about behaviour:**
How does it drape, fold, move?

This is the 6 step workflow we'll be working with today

The environments

- Underwater
- Extreme heat / desert
- Zero gravity / space



The environments are: underwater, extreme heat and zero gravity/space

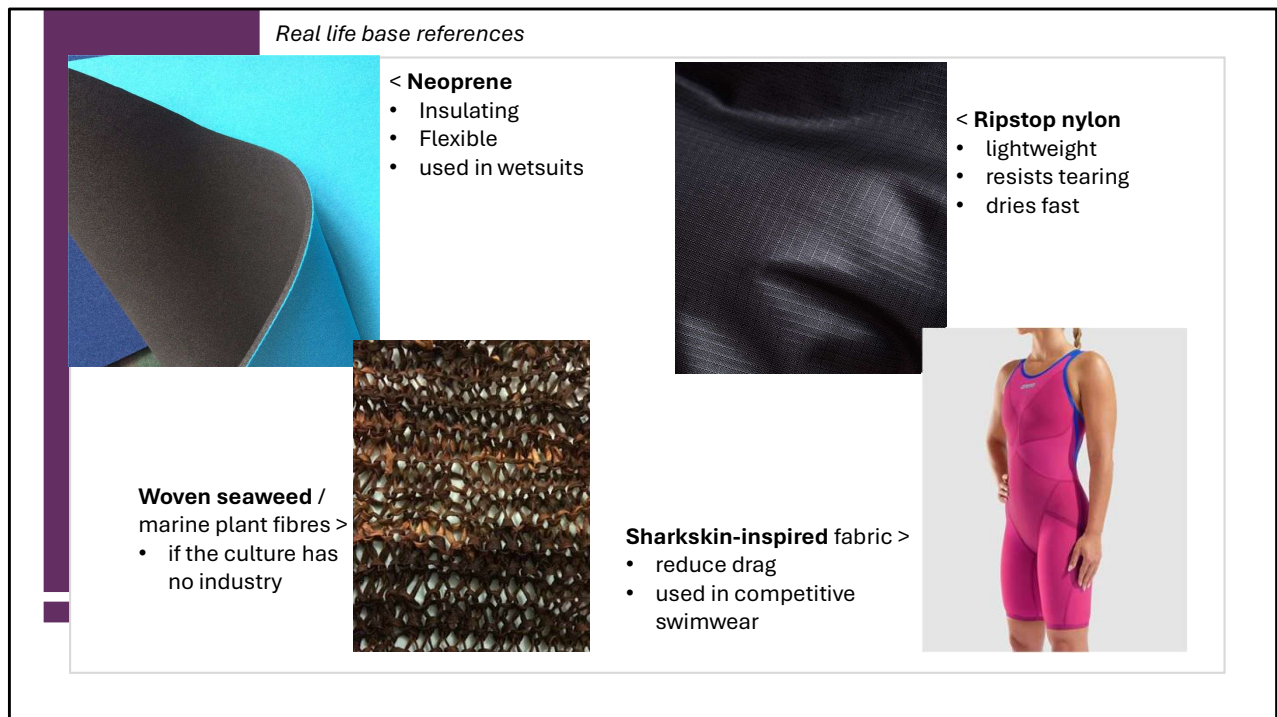
Underwater

The unique challenge:

- Water pressure increases with depth
- Constant moisture > nothing stays dry
- Movement through water requires minimal drag
- Temperature regulation works differently when submerged



Underwater gives us some unique challenges



Neoprene — insulating, flexible, used in wetsuits

Ripstop nylon — lightweight, resists tearing, dries fast

Woven seaweed / marine plant fibres — if the culture has no industry

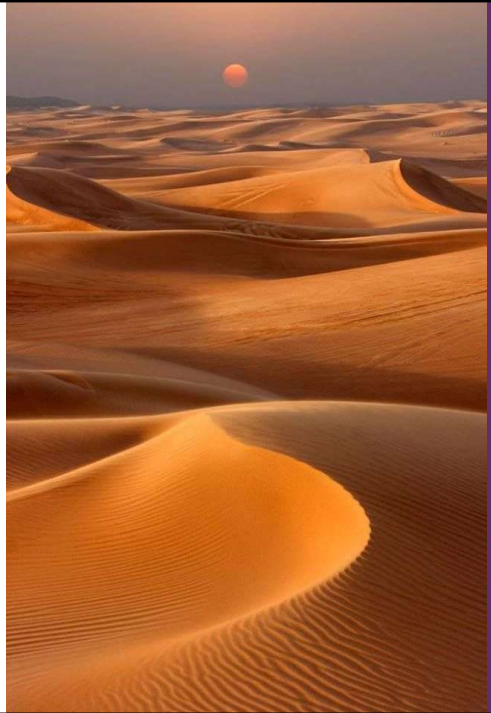
Sharkskin-inspired weaves — reduce drag (used in competitive swimwear)

The interesting design tension here is that protection and mobility fight each other underwater. More coverage means more drag. Thicker material means more insulation but less flexibility. Every choice is a trade-off. Worth asking — does this character swim or just live near water? That changes everything.

Extreme heat / desert

The unique challenge:

- Radiant heat from above AND reflected heat from ground
- Sand and dust get into everything > abrasion is constant
- Dehydration means sweat is precious > fabric needs to manage moisture carefully
- UV exposure is extreme



The unique challenges of the desert are:



Linen — breathable, moisture-wicking, used in hot climates for thousands of years

Merino wool — counterintuitive but regulates temperature in both heat and cold

Reflective Mylar — used in emergency blankets, reflects radiant heat

Nomex — fire and heat resistant, used in firefighter suits and racing suits

Tightly woven cotton — blocks UV and sand while staying breathable

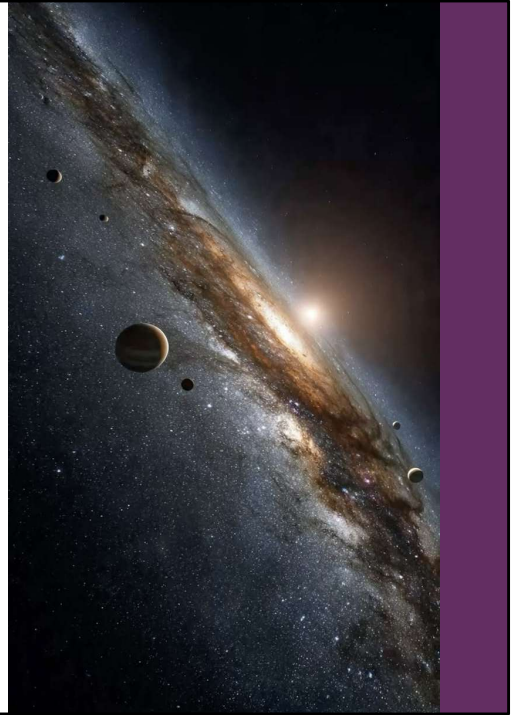
The desert is interesting because the instinct is to wear as little as possible but that's actually wrong. Covering up protects from radiant heat, UV and sand.

Traditional desert cultures figured this out thousands of years ago — long loose layers are the solution, not bare skin. That real-world logic should inform any speculative material designed for this environment.

Zero gravity / space

The unique challenge:

- No gravity means fabric floats and doesn't drape, it just... hovers
- Temperature swings are extreme
(from +120°C in sunlight to -160°C in shadow)
- Micrometeoroid impacts are a real threat
- Pressure differential between suit interior and vacuum outside
- No atmosphere means no convection cooling



And the unique challenges of space are...

Real life base references



Extravehicular Mobility Unit Material Layout

Labels in the diagram include: Thermal micrometeoroid garment cover (ortho-fabric), Thermal micrometeoroid garment liner (multiple layers), Thermal micrometeoroid garment liner (neoprene-coated nylon ripstop), Pressure garment bladder (urethane-coated nylon), Neoprene (Dacron), LCVG Liner (Nylon Tissue), LCVG Water Transport Tubing, LCVG Outer Layer (Nylon-Spandex), Pressure Garment Bladder (Urethane-Coated Nylon), Restrainer (Dacron), TMG Liner (Neoprene-coated nylon ripstop), TMG Installation (Bubble layer insulation, ABL), and TMG Cover (Dyneema fabric).

< **Ortho-Fabric outer layer:** woven blend of Gore-Tex, Kevlar and Nomex, handles abrasion, tearing and micrometeoroid impacts

- **Aluminised Mylar:** multiple insulating layers reflecting heat back in
- **Spandex cooling underlayer:** water tubes woven in to actively cool the body
- **Neoprene-coated ripstop nylon:** additional micrometeoroid protection

The Axiom/Prada AxEMU suit (2024): next generation, designed for the lunar south pole >



the modern spacesuit is already a masterclass:

Ortho-Fabric outer layer — woven blend of Gore-Tex, Kevlar and Nomex, handles abrasion, tearing and micrometeoroid impacts

Aluminised Mylar — multiple insulating layers reflecting heat back in

Spandex cooling underlayer — water tubes woven in to actively cool the body

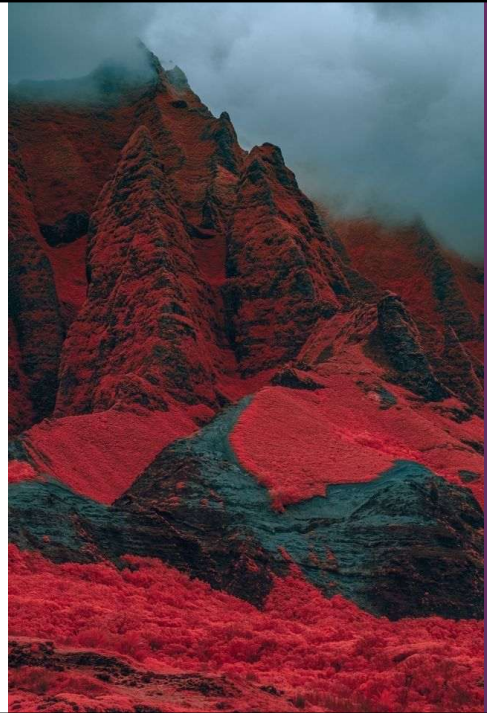
Neoprene-coated ripstop nylon — additional micrometeoroid protection

The Axiom/Prada AxEMU suit (2024) — next generation, designed for the lunar south pole

The spacesuit is genuinely one of the most sophisticated textile constructions humans have ever made. 14 layers, each doing a specific job. Worth showing a cross-section image if you can find one. The Prada collaboration is also a fun detail — high fashion meeting extreme engineering. And the big thing about zero gravity that students might not have thought about: fabric doesn't drape. Gravity is what makes cloth sim work the way it does. Take that away and all your assumptions about how clothing looks and moves are gone.

Quick question

- Which environment would be the hardest to design for?
- And which one connects most to something you're currently working on?





Activity

Section 2



The activity

- Design a speculative material using AI.
- Using the prompt template
- Then fill in the blanks
- Paste it into Claude
- Let it ask you follow-up questions
- Then it generates your design brief.



We're going to use AI to design a brief for a speculative material
So log into Claude using your school account and let's get started

AI prompt

- Posted in chat
- Choose your current character for this or think of something new while filling in the blanks



```
I am designing a speculative material for a game character.  
Help me develop a material design brief by first asking me  
3-5 follow-up questions to better understand my idea.  
Then, based on my answers, generate a structured design brief.
```

```
Here is my starting point:
```

```
Character/species: [who is wearing this]  
Environment: [underwater / extreme heat / zero gravity / other]  
Purpose of the material: [what problem does it need to solve]  
Cultural context: [what kind of society or culture made this]  
Technology level: [primitive / industrial / near future / far future]  
One real-life material I want to use as a base: [e.g. neoprene, Kevlar, linen]
```

```
Ask me your follow-up questions before generating the brief.
```

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Help me develop a material design brief by first asking me
3-5 follow-up questions to better understand my idea.
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Here is my starting point:

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Ask me your follow-up questions before generating the brief.

Key takeaways

- Every extreme environment breaks a different assumption about what clothing does
- Real materials are your starting point, not your ceiling
- Limitations make speculative materials believable
- Gravity is what makes fabric drape > designing for zero-g means rethinking everything







Show your work!

- Guidelines for sharing
- Discussion/feedback framework

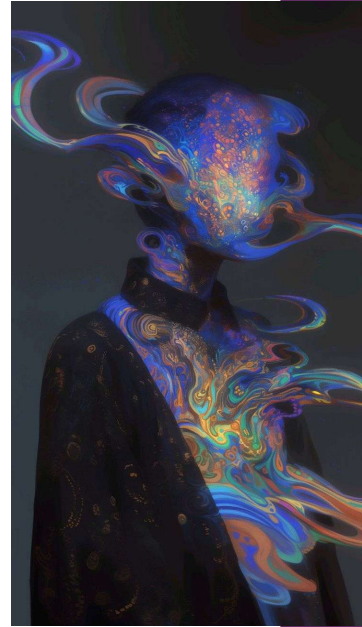


Reflection / wrap-up

Section 4

Reflection prompt

- What will you apply?
- Or take with you?



Thank you!!

- My last guild
- Thank you for joining
- You've helped me graduate
- I hope you've learned a lot and you'll take this with you in coming projects



This is the last guild.

It's been a genuine pleasure designing these sessions, watching you all engage with costume in ways I didn't expect, and collecting all of it as part of my research. You've helped me graduate.

I hope something from these guilds sticks, even just one thing you'll think about next time you put a character in an outfit.

Thank you for showing up every Friday.